Sequential Monitoring for Structural Breaks in Heterogeneous Panels: Design, Derivations, and Computational Details

Abstract

This note describes a Monte Carlo study of a real-time structural break monitoring procedure for heterogeneous panels. The procedure works in three steps: (1) each series is pre-filtered by an autoregression estimated on a calibration sample; (2) the filtered residuals are aggregated across series via $N(N-1)$ separate (unpooled) pairwise regressions, one OLS fit per ordered pair of series; (3) the cumulative sum of squares (CUSQ) of the aggregate residual is compared period by period against a boundary derived from the Brownian bridge, following Wang and Hsiao (2013). The note gives the exact data generating processes, a closed-form derivation showing that the $N(N-1)$-pair aggregation, even though every pair has its own fitted slope, can be evaluated exactly as a weighted cross-sectional average without ever materializing the $N(N-1)$ residual series, the two-scenario Monte Carlo design (size under no break, power and detection delay under a known break), and two computational improvements — FFT-based fractional differencing and vectorized AR residuals — that make the full 1.9×10^6-replication design feasible without changing any statistic. A companion design adds four targeted power experiments — common-factor and subgroup-specific volatility breaks, and common-factor and subgroup-specific loading breaks — built on the same AR-filtering, pairwise-aggregation, and CUSQ machinery.

1 Setup

We observe a panel of N time series $\{Y_{i,t}\}$ for $i = 1, \ldots, N$ and $t = 1, \ldots, T$. The first M periods form the calibration sample, assumed to be free of breaks. From period $M + 1$ onward the monitor checks, one period at a time, whether the panel has shifted.

The null and alternative hypotheses are

$$H_0: \text{ no break for any } t > M,$$
$$H_1: \text{ all } N \text{ series shift by } \delta \neq 0 \text{ at some time } \tau > M.$$

Sections 2–5 describe the four steps of the procedure: data generation, AR filtering, cross-sectional aggregation, and the CUSQ test. Section 7 describes the Monte Carlo design, Section 8 describes four additional targeted power experiments built on the same monitoring procedure, and Section 9 documents the computational improvements.

2 Data Generating Processes

2.1 Panel series

Each series follows an ARFIMA(p_i, d_i, q_i) model,

$$\Phi_i(L)(1-L)^{d_i}Y_{i,t} = \Theta_i(L)\,\varepsilon_{i,t}, \qquad \varepsilon_{i,t} \sim \text{i.i.d. } N(0,1),$$

where $\Phi_i(L) = 1-\phi_i L$ and $\Theta_i(L) = 1+\theta_i L$ are at most first-order polynomials and $(1-L)^{d_i}$ is the fractional differencing operator. Two panel sizes are used.

$N = 10$. Five pairs of series, each pair sharing the same parameters (Table 1).

Table 1: DGP for $N = 10$.

Series	Model	ϕ_i	d_i	θ_i
$1, 2$	ARMA$(1, 1)$	0.8	0	0.5
$3, 4$	ARMA$(1, 1)$	0.5	0	0.3
$5, 6$	ARFIMA$(0, d, 0)$	0	0.4	0
$7, 8$	ARFIMA$(1, d, 1)$	0.1	0.3	0.8
$9, 10$	ARFIMA$(1, d, 0)$	0.7	0.4	0

$N = 30$. A random-coefficient panel divided into four blocks (Table 2). The coefficients are drawn independently for each series and for each Monte Carlo replication. This means the simulation averages over both the randomness in the coefficients and the randomness in the innovations, giving a measure of average performance across the whole class of DGPs described in the table rather than performance at one fixed parameter vector.

Table 2: DGP for $N = 30$ (random coefficients, redrawn each replication).

Series	Model	ϕ_i	d_i	θ_i
1–15	ARMA$(1, 1)$	$U(0,\ 0.5)$	0	0.6
16–20	ARMA$(1, 1)$	$U(0.7,\ 0.9)$	0	0.6
21–25	ARFIMA$(0, d, 0)$	0	$U(0,\ 0.5)$	0
26–30	ARFIMA$(1, d, 1)$	$U(0,\ 0.4)$	$U(0,\ 0.5)$	$U(0,\ 0.3)$

2.2 Common factors

Four common factors are generated independently of the panel series (Table 3). They cover i.i.d. noise, a near-unit-root AR(1), and two long-memory processes.

Table 3: Common factor DGPs.

Factor	Model	ϕ	d
1	i.i.d. $N(0,1)$	0	0
2	AR(1)	0.99	0
3	ARFIMA$(0,d,0)$	0	0.20
4	ARFIMA$(0,d,0)$	0	0.45

2.3 Observed data and break injection

The observed series add a common factor with loading $\lambda = 0.5$:

$$Y^*_{i,t} = Y_{i,t} + 0.5\, f_t, \qquad i = 1, \ldots, N. \tag{1}$$

Under H_1, a level shift of size δ is added at time τ:

$$Y^{**}_{i,t} = Y^*_{i,t} + \delta \cdot \mathbf{1}\{t \geq \tau\}, \qquad i = 1, \ldots, N. \tag{2}$$

The break time τ is specified in Section 7.2.

3 AR Pre-Filtering

For each series i, we fit an AR(p) model on the calibration sample $t = 1, \ldots, M$, where

$$p = \max\left(3,\ 3\left\lfloor \frac{M-1}{100} \right\rfloor\right). \tag{3}$$

The OLS estimate from the calibration sample is

$$\hat{\beta}_{AR} = \arg\min_{b \in \mathbb{R}^p} \sum_{t=p+1}^{M} \Big(Y_{i,t} - b'(Y_{i,t-1}, \ldots, Y_{i,t-p})'\Big)^2. \tag{4}$$

This estimate is fixed after period M and never updated. For all $t = p+1, \ldots, T$, the one-step-ahead residual is

$$\hat{e}_{i,t} = Y_{i,t} - \hat{\beta}'_{AR}(Y_{i,t-1}, \ldots, Y_{i,t-p})'. \tag{5}$$

The first seven observations are discarded as a burn-in. Fixing the AR coefficients after calibration is the key sequential monitoring requirement: parameters are estimated from clean data and then held constant, so the monitoring period residuals are not used to update the model.

4 Cross-Sectional Aggregation

4.1 Unpooled pairwise regression

Let $\hat{e}_{i,t}$ denote the AR residuals for $i = 1, \ldots, N$ and $t = 1, \ldots, T$. For every ordered pair (i, j) with $i \neq j$ (there are $N(N-1)$ such pairs), we fit a *separate* calibration-period regression

$$\hat{e}_{i,t} = \alpha_{ij} + \beta_{ij}\, \hat{e}_{j,t} + u_{i,j,t}, \qquad t = 1, \ldots, M. \tag{6}$$

Unlike a pooled specification, the $N(N-1)$ regressions are *not* stacked into a single sample: each pair (i, j) has its own OLS estimate $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$, computed using only that pair's M calibration observations. As with the AR coefficients of Section 3, $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$ are estimated once from the calibration sample and then held fixed for every $t = 1, \ldots, T$. For all t, the residual from pair (i, j)'s regression is

$$U_{i,j,t} = \hat{e}_{i,t} - \hat{\alpha}_{ij} - \hat{\beta}_{ij}\, \hat{e}_{j,t}. \tag{7}$$

The aggregate signal used in the CUSQ test is, as before, the average of $U_{i,j,t}$ over all ordered pairs:

$$\bar{e}_t \equiv \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^{N} U_{i,j,t}, \qquad t = 1, \ldots, T. \tag{8}$$

As stated, computing (8) requires fitting all $N(N-1)$ pairs and forming all $N(N-1)$ residual sequences — an $O(N^2 M)$ estimation step followed by an $O(N^2 T)$ evaluation step. The next two propositions show that every $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$, and hence $\bar{e}_t$, can be obtained in closed form at a total cost of $O(N^2 M)$ (estimation) plus $O(NT)$ (evaluation), without ever materializing the $N(N-1)$ residual series and without any approximation.

4.2 Closed form for $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$

Define the per-series calibration mean

$$\mu_i \equiv \frac{1}{M} \sum_{t=1}^{M} \hat{e}_{i,t}, \qquad i = 1, \ldots, N, \tag{9}$$

and the demeaned residual, held fixed and applied to *every* t (not only the calibration sample),

$$f_{i,t} \equiv \hat{e}_{i,t} - \mu_i, \qquad t = 1, \ldots, T. \tag{10}$$

Let $d_{i,t} \equiv f_{i,t}$ for $t \leq M$ denote the calibration-period values, and define the $N \times N$ Gram matrix of calibration-period centered residuals

$$S_{ij} \equiv \sum_{t=1}^{M} d_{i,t}\, d_{j,t}, \qquad i, j = 1, \ldots, N. \tag{11}$$

Proposition 1. *The single-pair OLS estimates from* (6) *are, for* $i \neq j$,

$$\hat{\beta}_{ij} = \frac{S_{ij}}{S_{jj}}, \qquad \hat{\alpha}_{ij} = \mu_i - \hat{\beta}_{ij}\,\mu_j. \tag{12}$$

Proof. For a fixed pair (i, j), (6) is an ordinary simple linear regression of $\hat{e}_{i,t}$ on $\hat{e}_{j,t}$ over $t = 1, \ldots, M$, with no pooling across pairs. Its OLS slope is the ratio of the sample covariance of $(\hat{e}_{i,t}, \hat{e}_{j,t})$ to the sample variance of $\hat{e}_{j,t}$: $\sum_t d_{i,t}d_{j,t} \big/ \sum_t d_{j,t}^2 = S_{ij}/S_{jj}$. The intercept follows from the standard simple-regression identity $\hat{\alpha}_{ij} = \mu_i - \hat{\beta}_{ij}\mu_j$. □

4.3 Closed form for $\bar{e}_t$

Define the cross-sectional sum of the centered residuals, $F_t \equiv \sum_{i=1}^{N} f_{i,t}$, and, for each j, the column sum of the fixed pairwise slopes excluding the diagonal,

$$c_j \equiv \sum_{i \neq j} \hat{\beta}_{ij}, \qquad j = 1, \ldots, N. \tag{13}$$

Proposition 2. *For every* $t = 1, \ldots, T$,

$$\bar{e}_t = \frac{(N-1)F_t - \displaystyle\sum_{j=1}^{N} c_j f_{j,t}}{N(N-1)}. \tag{14}$$

Proof. By (12), $\hat{\alpha}_{ij} + \hat{\beta}_{ij}\mu_j = \mu_i$, so $U_{i,j,t} = \hat{e}_{i,t} - \hat{\alpha}_{ij} - \hat{\beta}_{ij}\hat{e}_{j,t} = f_{i,t} - \hat{\beta}_{ij} f_{j,t}$. Expanding (8):

$$\bar{e}_t = \frac{1}{N(N-1)} \sum_{i \neq j} \bigl(f_{i,t} - \hat{\beta}_{ij} f_{j,t}\bigr) = \frac{1}{N(N-1)} \left[\sum_{i \neq j} f_{i,t} - \sum_{i \neq j} \hat{\beta}_{ij} f_{j,t} \right].$$

In the first sum, $f_{i,t}$ is added once for every $j \neq i$, i.e. $N-1$ times for each i, so $\sum_{i \neq j} f_{i,t} = (N-1)F_t$. In the second sum, regrouping by j gives $\sum_{i \neq j} \hat{\beta}_{ij} f_{j,t} = \sum_j f_{j,t} \sum_{i \neq j} \hat{\beta}_{ij} = \sum_j c_j f_{j,t}$. Combining the two terms gives (14). □

Remark 1. *Unlike under pooling, the* $N(N-1)$ *pairwise regressions here do* not *share a common slope, so* $\bar{e}_t$ *is no longer a simple affine map of the plain cross-sectional mean: each series* j *contributes through* $f_{j,t}$ *with its own pair-specific weight* c_j*.* (14) *is nonetheless exact and never requires forming the* $N(N-1)$ *residual series. The cost is* $O(N^2M)$ *to form* S *(and hence every* $\hat{\beta}_{ij}$ *and* c_j*) once from the calibration sample, plus* $O(NT)$ *to evaluate* $\bar{e}_t$ *for the full sample — versus* $O(N^2T)$ *for a direct, pair-by-pair evaluation. For* $N = 30$ *this still replaces an* $N(N-1) = 870$*-pair construction with one* 30×30 *Gram matrix and a single weighted cross-sectional average.*

5 The CUSQ Statistic and Monitoring Boundary

5.1 Statistic

Let $C_l = \sum_{t=1}^{l} \bar{e}_t^2$ be the cumulative sum of squares of $\bar{e}_t$. Following Wang and Hsiao (2013) (p. 5), the centered CUSQ for $n \geq m$ is

$$D_n \equiv \frac{C_n}{C_m} - \frac{n-2}{m-2}, \tag{15}$$

and the detector is

$$T_n \equiv \sqrt{\frac{m}{2}}\, D_n = \sqrt{\frac{m}{2}} \left(\frac{C_n}{C_m} - \frac{n-2}{m-2} \right). \tag{16}$$

This is equivalent to $T_n = m^{-1/2} \hat{S}_n$ in the notation of Wang and Hsiao (2013), where $\hat{S}_n = 2^{-1/2} m \hat{D}_n$.

5.2 Limiting distribution

Under H_0, Wang and Hsiao (2013) (Theorem 1) show that

$$T_n = m^{-1/2} \hat{S}_{[m\tau]} \;\Rightarrow\; W^0(\tau), \qquad \tau = \frac{n}{m} \geq 1, \tag{17}$$

where $W^0(\tau) = W(\tau) - \tau W(1)$ is a Brownian bridge and $W(\tau)$ is a standard Wiener process.

5.3 Monitoring boundary

Following Chu et al. (1996), the crossing probability of $W^0(\tau)$ satisfies (Wang and Hsiao, 2013, eq. (5)):

$$P\big\{ |W^0(\tau)| \geq (\tau-1)^{-1} [\tau(a^2 + \ln \tau)]^{1/2} \text{ for some } \tau \geq 1 \big\} = 2[1 - \Phi(a)] + a\phi(a), \tag{18}$$

where Φ and ϕ are the standard-normal CDF and PDF. Setting the right-hand side equal to α and solving numerically gives

$$a^2_{10\%} = 6.2514, \qquad a^2_{5\%} = 7.8147, \qquad a^2_{1\%} = 11.3449. \tag{19}$$

Wang and Hsiao (2013) state $a^2_{5\%} = 7.78$ (rounded to two decimal places). Discretising λ as n/m yields the boundary function (Wang and Hsiao, 2013, p. 6):

$$g\Big(\frac{n}{m};\, a^2_\alpha\Big) = \frac{n-m}{m} \left[\frac{n}{n-m} \left(a^2_\alpha + \ln \frac{n}{n-m} \right) \right]^{1/2}. \tag{20}$$

6 Sequential Decision Rule

At each period $n > m$, compute T_n from (16) and compare it to the boundary $g(n/m;\, a_\alpha^2)$ from (20). The procedure declares a break at the first period where the statistic crosses the boundary:

$$\hat{\tau}_\alpha \equiv \inf\bigl\{n > m : |T_n| \ge g\bigl(n/m;\, a_\alpha^2\bigr)\bigr\}, \tag{21}$$

with $\hat{\tau}_\alpha = +\infty$ if no crossing occurs before T. The AR coefficients $\hat{\beta}_{AR}$, the pairwise regression coefficients $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$ for every $i \neq j$, and the per-series means μ_i are all fixed from the calibration period and never updated after $t = m$.

7 Monte Carlo Design

The total sample length used in each replication is

$$T_{\text{total}} = 10M + 10, \tag{22}$$

which provides a monitoring window of $9M$ periods beyond the calibration sample. Two scenarios are run.

7.1 Scenario A: size ($\delta = 0$)

No break is injected. Because the data do not depend on k, each simulated path yields one stopping time $\hat{\tau}_\alpha^{(r)}$, and this is scored against six window lengths $k \in \mathcal{K} = \{0.2, 0.3, 0.5, 1, 2, 3\}$ at once:

$$A_\alpha^{(r)}(k) \equiv \mathbf{1}\bigl\{\hat{\tau}_\alpha^{(r)} \le M + kM\bigr\}. \tag{23}$$

The empirical size is

$$\widehat{\text{Size}}_\alpha(N, M, \text{factor}, k) = \frac{1}{R}\sum_{r=1}^{R} A_\alpha^{(r)}(k), \tag{24}$$

which estimates the false alarm probability within the first kM monitoring periods. A well-calibrated test should give $\widehat{\text{Size}}_\alpha \approx \alpha$.

7.2 Scenario B: power ($\delta \in \{1, 2, 3\}$)

A break of size δ is injected at

$$\tau(k) = M + \lfloor kM \rfloor, \qquad k \in \mathcal{K}. \tag{25}$$

Here k is the break location, not a scoring window. Each (δ, k) combination requires its own set of replications because the data depend on (δ, k). The stopping time $\hat{\tau}_\alpha^{(r)}$ is compared to the true break date $\tau(k)$ and classified as one of three outcomes:

$$P_\alpha^{(r)} \equiv \mathbf{1}\{\hat{\tau}_\alpha^{(r)} \le \tau(k)\} \qquad \text{(alarm before the break)}, \tag{26}$$

$$D_\alpha^{(r)} \equiv \mathbf{1}\{\tau(k) < \hat{\tau}_\alpha^{(r)} < \infty\} \qquad \text{(alarm after the break)}, \tag{27}$$

$$B_\alpha^{(r)} \equiv P_\alpha^{(r)} + D_\alpha^{(r)} = \mathbf{1}\{\hat{\tau}_\alpha^{(r)} < \infty\} \qquad \text{(any alarm)}. \tag{28}$$

The Monte Carlo averages are

$$\widehat{P}_\alpha = \frac{1}{R}\sum_r P_\alpha^{(r)}, \qquad \widehat{D}_\alpha = \frac{1}{R}\sum_r D_\alpha^{(r)}, \qquad \widehat{B}_\alpha = \widehat{P}_\alpha + \widehat{D}_\alpha. \tag{29}$$

$\widehat{D}_\alpha$ is the power of the test — the probability of detecting the break after it occurs. $\widehat{P}_\alpha$ is the premature alarm rate.

For replications where $D_\alpha^{(r)} = 1$, the detection delay is

$$\text{Delay}_\alpha^{(r)} \equiv \hat{\tau}_\alpha^{(r)} - \tau(k) > 0. \tag{30}$$

Summary statistics (mean, standard deviation, median, 25th and 75th percentiles) are computed over the subset $\{r : D_\alpha^{(r)} = 1\}$. Premature alarms ($P_\alpha^{(r)} = 1$) are excluded from this calculation because they have a negative delay and measure a different phenomenon.

7.3 Full design

Table 4 lists all parameter combinations.

Table 4: Monte Carlo design.

Parameter	Scenario A (size)	Scenario B (power)
N	$\{10, 30\}$	$\{10, 30\}$
δ	0	$\{1, 2, 3\}$
k	scoring window, $\{0.2, 0.3, 0.5, 1, 2, 3\}$	break location, $\{0.2, 0.3, 0.5, 1, 2, 3\}$
M	$\{50, 80, 100, 200, 300\}$	$\{50, 80, 100, 200, 300\}$
Factor	$\{1, 2, 3, 4\}$	$\{1, 2, 3, 4\}$
R	2500	2500
Combinations	$2 \times 5 \times 4 = 40$	$2 \times 3 \times 6 \times 5 \times 4 = 720$
Total replications	10^5	1.8×10^6

For $N = 30$, the random coefficients in Table 2 are redrawn on every replication, so the Monte Carlo averages integrate over both the coefficient distribution and the innovation sequences.

8 Targeted Power Experiments

The design of Section 7 injects a level shift δ directly into every series. A companion design instead breaks the panel *indirectly*, through the common factor that loads onto the series, and asks whether the monitor still has power when the break works through the factor structure rather than through a direct level shift. Four experiments are run, crossing two break mechanisms (a break in the factor's *volatility* versus a break in its *loading*) with two factor structures (a single factor shared by the whole panel versus factor-specific subgroups). All four reuse, unchanged, the AR pre-filtering of Section 3, the pairwise cross-sectional aggregation (14), and the CUSQ statistic and boundary of Section 5: only the data-generating process and the reported outcomes differ. The AR coefficients $\hat{\beta}_{AR}$, the pairwise coefficients $(\hat{\alpha}_{ij}, \hat{\beta}_{ij})$, and the means μ_i are again estimated once on $t = 1, \ldots, M$ and held fixed thereafter. In all four experiments, k no longer plays the dual role of Section 7 (scoring window in Scenario A, break location in Scenario B): here k is only a break location, restricted to $k \in \{0.2, 0.3, 0.5\}$, and the break time is $\tau^*(k) = M + \lfloor kM \rfloor$ as in (25). Outcomes are reported with the alarm/premature/detect/delay logic of (26)–(30), with $\tau(k)$ replaced by $\tau^*(k)$; to match the implementation's column names, the Monte Carlo averages $\widehat{B}_\alpha$, $\widehat{P}_\alpha$, $\widehat{D}_\alpha$ are denoted AR_α (alarm rate), PAR_α (premature alarm rate), and DR_α (detection rate), reported alongside the same five delay statistics. $R = 2500$ replications are run per combination, exactly as in Section 7.

8.1 Experiment 1: common-factor volatility break

Every series loads on factor 4 (the long-memory ARFIMA$(0, 0.45, 0)$ factor of Table 3) with the same fixed loading as (1), $\gamma = 0.5$. No level shift is injected; instead, the i.i.d. $N(0,1)$ innovations that drive factor 4 through fractional differencing, (34)–(35), switch standard deviation from 1 to $\sqrt{3}$ at $\tau^*(k)$:

$$\varepsilon_{4,t} \sim N(0,1) \text{ for } t < \tau^*(k), \qquad \varepsilon_{4,t} \sim N(0,3) \text{ for } t \geq \tau^*(k). \tag{31}$$

The idiosyncratic panel (Tables 1–2) is unaffected; only the variance of the common shock changes, so the break reaches every series solely through the factor loading.

8.2 Experiment 2: common-factor loading break

Each factor $j \in \{1, 2, 3, 4\}$ is now considered separately (no volatility break: the factor itself is generated exactly as in Table 3 throughout). Instead, every series' loading on $f_{j,t}$ jumps from $\gamma_{\text{pre}} = 0.1$ to $\gamma_{\text{post}} = 0.8$ at $\tau^*(k)$:

$$Y^*_{i,t} = Y_{i,t} + \gamma(t)\, f_{j,t}, \qquad \gamma(t) = \begin{cases} 0.1, & t < \tau^*(k) \\ 0.8, & t \geq \tau^*(k), \end{cases} \qquad i = 1, \ldots, N. \tag{32}$$

Because each of the four factors is run as a separate combination, the parameter grid additionally ranges over factor $\in \{1, 2, 3, 4\}$, as in Scenario B of Section 7.

8.3 Experiments 3.1–3.2: heterogeneous (subgroup) factor structure

Experiments 8.1–8.2 load every series on the same single factor. Experiments 3.1 and 3.2 instead partition the N series into three disjoint subgroups G_2, G_3, G_4, each loading on a *different* factor (f_2, f_3, f_4 respectively) and on no other factor (Table 5). Because no factor is shared by the whole panel, the panel-wide pooled signal is, by construction, only partially informative about a break confined to a single subgroup — this is the sense in which the factor structure is "weak" relative to Experiments 8.1–8.2.

Table 5: Subgroup composition for Experiments 3.1–3.2.

	Factor	$N = 10$	$N = 30$
G_2	f_2	$\{s_1, s_2, s_3\}$	$\{s_1, \ldots, s_3\}$
G_3	f_3	$\{s_4, s_5, s_6\}$	$\{s_4, \ldots, s_{12}\}$
G_4	f_4	$\{s_7, \ldots, s_{10}\}$	$\{s_{13}, \ldots, s_{30}\}$

For every replication, AR filtering is applied to the whole panel exactly as in Section 3 — each series keeps its own fixed AR(p) coefficients regardless of subgroup membership. Two CUSQ tests are then computed from the *same* AR residuals: a full-panel test using (14) with all N columns, and, for each $g \in \{2, 3, 4\}$, a subgroup test using (14) restricted to the $|G_g|$ columns in G_g only. The closed form of Proposition 2 applies verbatim to any subset of columns, so the subgroup tests require no additional derivation; only the cross-sectional aggregation step changes, never the AR filtering. This lets each replication compare whole-panel monitoring against subgroup-specific monitoring for a break confined to one subgroup.

Experiment 3.1 (volatility break). Each subgroup's own factor receives the same volatility break as (31) (std. $1 \to \sqrt{3}$ at $\tau^*(k)$), with a fixed loading $\gamma = 0.5$, applied only to the columns in that subgroup.

Experiment 3.2 (loading break). Each subgroup's own factor has constant volatility, but the subgroup's loading on it jumps from a subgroup-specific γ_{pre} to γ_{post} at $\tau^*(k)$ (Table 6), analogous to (32) but applied subgroup-by-subgroup with its own factor.

Table 6: Pre- and post-break loadings for Experiment 3.2.

Subgroup	Factor	γ_{pre}	γ_{post}
G_2	f_2	0.10	0.60
G_3	f_3	0.25	0.85
G_4	f_4	0.30	0.90

For both 3.1 and 3.2, the full-panel metrics are reported with prefix "full_" and each subgroup g's metrics with prefix "gg_". In addition, for every pair of subgroups $(a, b) \in \{(2, 3), (3, 4), (2, 4)\}$ the gap in 5%-level detection rate is reported,

$$\Delta\text{DR}_{g_a, g_b} \equiv \text{DR}_{g_a, 5\%} - \text{DR}_{g_b, 5\%}, \tag{33}$$

summarizing whether a break carried by a more persistent (higher-d or higher-ϕ) factor is detected more or less reliably than one carried by a less persistent factor.

8.4 Design summary

Table 7 lists the four experiments' combinations. M, factor, and the $N = 30$ random-coefficient redraw rule are exactly as in Table 4.

Table 7: Targeted power experiment design.

Experiment	N	M	k	Factor
1 (factor-4 vol. break)	$\{10, 30\}$	$\{50, 80, 100, 200, 300\}$	$\{0.2, 0.3, 0.5\}$	—
2 (loading break)	$\{10, 30\}$	$\{50, 80, 100, 200, 300\}$	$\{0.2, 0.3, 0.5\}$	$\{1, 2, 3, 4\}$
3.1 (subgroup vol. break)	$\{10, 30\}$	$\{50, 80, 100, 200, 300\}$	$\{0.2, 0.3, 0.5\}$	—
3.2 (subgroup loading break)	$\{10, 30\}$	$\{50, 80, 100, 200, 300\}$	$\{0.2, 0.3, 0.5\}$	—
Combinations: $30 + 120 + 30 + 30 = 210$; $R = 2500$ each; total replications $= 5.25 \times 10^5$.				

9 Computational Details

The full design requires 1.9×10^6 replications. Two changes to the implementation reduce computation time substantially without changing any statistic.

9.1 FFT-based fractional differencing

An ARFIMA process requires applying the fractional integration operator $(1 - L)^{-d}$, which has the moving-average representation

$$(1 - L)^{-d} = \sum_{j=0}^{\infty} w_j L^j, \qquad w_j = \frac{\Gamma(j + d)}{\Gamma(j + 1)\,\Gamma(d)}. \tag{34}$$

Given a short-memory series Y_t, the fractionally integrated series is

$$X_t = \sum_{j=0}^{t} w_j Y_{t-j}, \qquad t = 0, 1, \ldots, T - 1. \tag{35}$$

Computing (35) by a direct loop costs $O(T^2)$ operations. The same result can be obtained by the discrete convolution

$$(w * Y) = \mathcal{F}^{-1}\big[\mathcal{F}(w) \odot \mathcal{F}(Y)\big], \tag{36}$$

where $\mathcal{F}$ is the DFT computed by the FFT algorithm, at cost $O(T \log T)$. The first T elements of $(w * Y)$ equal (35) exactly. For numerical stability, the weights are computed in log space:

$$w_j = \exp\left[\ln \Gamma(j+d) - \ln \Gamma(j+1) - \ln \Gamma(d)\right], \tag{37}$$

which avoids overflow for large j. For $T = 4000$, this reduces the fractional differencing step from hundreds of milliseconds to under one millisecond per series.

9.2 Vectorized AR residuals

Equation (5) computes one residual per time period. A direct implementation loops over t and computes a dot product at each step, incurring $T - p$ separate Python function call overheads. The same result follows from a single matrix multiplication.

Lemma 1. *Let $t_0 = \max(8, p+1)$ and $\mathcal{I} = \{t_0, \ldots, T\}$. Define the lag matrix $Z \in \mathbb{R}^{|\mathcal{I}| \times p}$ by*

$$Z_{t,\ell} = Y_{i,\,t-\ell}, \qquad \ell = 1, \ldots, p, \quad t \in \mathcal{I}, \tag{38}$$

and let $y = (Y_{i,t})_{t \in \mathcal{I}}$. Then

$$(\hat{e}_{i,t})_{t \in \mathcal{I}} = y - Z\hat{\beta}_{AR}. \tag{39}$$

Proof. Row t of $Z\hat{\beta}_{AR}$ equals $\sum_{\ell=1}^{p} \hat{\beta}_{AR,\ell} Y_{i,t-\ell}$, which is the predicted value in (5). So $(y - Z\hat{\beta}_{AR})_t = \hat{e}_{i,t}$ for every $t \in \mathcal{I}$. $\square$

The matrix Z is formed once and the residuals for all periods are obtained in a single BLAS call. For $T \approx 3000$ and $p \leq 6$, this is about 30 times faster than the loop version, with identical numerical output.

9.3 Parallelisation

After both improvements, the time for one replication at $N = 30$, $M = 300$ falls from roughly 0.14 seconds to roughly 0.02 seconds, a reduction of about $8\times$. Across 1.9×10^6 replications, this brings the projected single-thread runtime from about 72 hours to about 9.5 hours.

The 760 parameter combinations in Table 4 are independent of one another, so each combination is assigned to one of 8 worker processes. Each worker is seeded from OS entropy to ensure independent random streams across processes. With 8-way parallelism, the expected wall-clock time is 1–1.5 hours.

The unpooled pairwise aggregation of Section 4 adds an $O(N^2 M)$ Gram-matrix step per replication that was not present under pooling. For the grid used here ($N \leq 30$, $M \leq 300$) this step is small relative to data generation and AR filtering, so the per-replication timings above are not expected to change materially, but they should be re-benchmarked after the implementation update rather than assumed.

10 Summary

Scenario A produces 240 rows (40 combinations $\times$ 6 values of k), each reporting $\widehat{\text{Size}}_\alpha$ for $\alpha \in \{10\%, 5\%, 1\%\}$. Scenario B produces 720 rows, each reporting $\widehat{B}_\alpha$, $\widehat{P}_\alpha$, $\widehat{D}_\alpha$, and the five delay statistics for each α.

In every replication, the CUSQ statistic uses $D_n = C_n/C_m - (n-2)/(m-2)$ from (15) and the boundary $g(n/m)$ from (20), both as specified in Wang and Hsiao (2013). Propositions 1 and 2 ensure that the (unpooled) cross-pair aggregate $\bar{e}_t$, with every pair (i, j) contributing its own fitted slope $\hat{\beta}_{ij}$, is nonetheless computed exactly, at cost $O(N^2M) + O(NT)$ instead of $O(N^2M) + O(N^2T)$ for a direct pair-by-pair evaluation. Lemma 1 and the FFT-based fractional-differencing step ensure that all residuals are numerically identical to a direct implementation.

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